

JOSEPH DENG

josephdeng317@gmail.com | [Portfolio](#) | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of British Columbia

Bachelor of Science in Computer Science and Physics, Co-op program — 91.2% Average

Vancouver, BC

Sep 2023 – Apr 2028

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript/TypeScript,

Web: HTML, CSS, React.js, Node.js, Next.js, Bootstrap, Selenium

Frameworks/Tools: AWS Lambda, Bedrock, OpenAI API, Git, Flask, Matplotlib, Selenium SQLite, Blender

TECHNICAL EXPERIENCE

Fullstack Developer Intern | CS Toolkit Inc.

May – Jun 2024

- Implemented dynamic API calls to Contentful using **Flask** to add pagination, tagging, and a search function to the company's **production blog page**.
- Engaged in a **multi-stage build and review process** with a professional UX designer, and reskinned the website to match a Figma prototype with every detail accounted for
- Followed industry best practices for **Git branching and pull requests** when contributing

Programmer, Designer, Builder | VEX Robotics Team 4659A

Sep 2022 – Mar 2023

- Worked on both hardware and software for a **provincial qualifying** VEX Spin Up Robot
- Helped implement an auto-aim system with **computer vision** to improve shooter accuracy by **200%**

TECHNICAL PROJECTS

Urbanize AI | AWS Lambda, Bedrock, Expo, React.js, Python | [Demo](#)

Oct 2024

- Created an **AI-powered Urban Design** Improvement app that helps users identify changes that could be made in their own neighborhoods to improve sustainability, placed **top 4 out of 25+ teams** at UBC CIC Hackathon
- Implemented a serverless **AWS Lambda** function that made calls to send text and base-64 representations of images to an **generative AI model through Bedrock**
- Debugged Lambda function with Postman and CloudWatch to allow for easier testing of the API

AI Recipe Generator | Next.js, Node.js, React.js, Three.js, OpenAI API, Bootstrap | [Live Website](#)

May – Aug 2024

- Implemented a **Node.js** backend that made asynchronous calls to the **OpenAI API** to create **AI-generated recipes** based on ingredients entered by the user.
- Designed the frontend web application with React to change the visibility of elements dynamically, and used Three.js to insert a custom interactive **3D model created using Blender**.
- Refactored project to use the **Next.js app router**, enabling a **serverless deployment with Vercel**

Musical Ear Trainer | Java, JUnit |

Jan – Apr 2024

- Built an interface with Java Swing that quizzes the user on randomly generated notes made audible with MidiSynth
- Created a logical class hierarchy for storing previous quiz results which allowed the user to write and read their data to and from **Json and SQLite**
- Designed a **comprehensive test suite using JUnit** to maintain a production-ready codebase.

ADDITIONAL WORK EXPERIENCE

Program Assistant | Shad Canada

Jun – Jul 2024

- Dedicated **90+** hour weeks to facilitate a STEAM program involving workshops and student-led entrepreneurship projects, displaying an unwavering sense of responsibility and commitment to student growth and innovation
- Maintained an engaged and respectful audience of **64** high school students during long hours of lectures & workshops, demonstrating strong leadership, patience, and communication skills
- Presented an interactive beginner-friendly workshop on 3D modeling using Blender.

Piano Teacher | Studio Vivian

May 2023 – Jun 2024

- Taught piano lessons to targeted towards youth and created **customized lesson plans** to cater to their individual interests and learning styles, was **consistently praised** by parents for keeping their kids engaged
- Engaged students by teaching them how to **learn popular music by ear**, helping them develop a better sense of musicianship and build their passion for the art